Computer Architecture

Lecture 2a: Course Info & Logistics

Prof. Onur Mutlu
ETH Zürich
Fall 2025
26 September 2025

Brief Self Introduction



- Onur Mutlu
  - Full Professor @ ETH Zurich ITET (INFK), since Sept 2015
  - Strecker Professor @ Carnegie Mellon University ECE (CS), 2009-2016, 2016-…
  - Started the Comp Arch Research Group @ Microsoft Research, 2006-2009
  - Worked @ Google, VMware, Microsoft Research, Intel, AMD, Stanford
  - PhD in Computer Engineering from University of Texas at Austin in 2006
  - BS in Computer Engineering & Psychology from University of Michigan in 2000
  - https://people.inf.ethz.ch/omutlu/ omutlu@gmail.com

- Research and Teaching in:
  - **Computer architecture, systems, hardware security, bioinformatics**
  - Memory and storage systems
  - Robust & dependable hardware systems: security, safety, predictability, reliability
  - Hardware/software cooperation
  - New computing paradigms; architectures with emerging technologies/devices
  - Architectures for bioinformatics, genomics, health, medicine, AI/ML
  - AI/ML for architecture and system design
  - …

My Co-Instructor



- Mohammad Sadrosadati
  - Senior Researcher and Lecturer @ SAFARI Research Group, ETHZ
  - PhD in Computer Engineering from Sharif University of Technology in 2019
  - MS in Computer Engineering from Sharif University of Technology in 2014
  - BS in Computer Engineering from Sharif University of Technology in 2012
  - mohammad.sadrosadati@safari.ethz.ch

- Research & Teaching Areas
  - Computer Architecture
  - Memory/Storage Systems
  - Near-Data Processing
  - Heterogeneous System Architecture
  - Bioinformatics
  - Interconnection Network

Our Head Teaching Assistant



- Konstantina Koliogeorgi
  - Senior Researcher and Lecturer @ SAFARI
  - PhD, National Technical University of Athens, 2023
  - kkoliogeorgi@safari.ethz.ch

- Research & Teaching Areas
  - Hardware/Software Co-Design
  - Heterogeneous System Architecture
  - Reconfigurable Computing and Architectures
  - Hardware Acceleration
  - Optimized Architectures for Genome analysis
  - High Level Synthesis Tools
  - Design Space Exploration

Course Info: The Teaching Team

- **Instructors**
  - **Professor Onur Mutlu**
  - **Dr. Mohammad Sadrosadati**
- **Teaching Assistants**
  - **Dr. Konstantina Koliogeorgi**
  - Dr. Zhiheng Yue
  - Rahul Bera
  - Konstantinos Kanellopoulos
  - Nika Mansouri Ghiasi
  - Ataberk Olgun
  - Rakesh Nadig
  - Nisa Bostanci
  - Ismail Emir Yüksel
  - Haocong Luo
  - Harshita Gupta
  - Mayank Kabra
  - Andreas Kosmas Kakolyris
  - Konstantinos Sgouras
  - Harsh Songara
  - Susana Rebolledo Ruiz
- **Get to know them and their research → Perhaps they can be your future mentors**

Some Goals of This Course

- Teach/enable/empower you to:
  - Understand tradeoffs in computer system design
  - Understand how decisions made in hardware affect the software/programmer as well as the hardware designer
  - Think critically (in solving problems)
  - Think broadly across the levels of transformation
  - Understand how to analyze and make tradeoffs in design
  - Apply the above in several lab projects, HWs, reviews

Major High-Level Goals of This Course

- Understand the principles and tradeoffs (of design)
- Understand the precedents

- Based on such understanding:
  - Enable you to evaluate tradeoffs of different designs and ideas
  - Enable you to develop principled designs
  - Enable you to develop novel, out-of-the-box designs

- The focus is on:
  - Principles, precedents, and how to use them for new designs

- In Computer Architecture

Why These Goals?

- Regardless of your future direction, learning the principles and methodical tradeoff analysis techniques we cover will be useful to
  - design better hardware
  - design better software
  - design better systems
  - make better tradeoffs in design
  - understand why computers behave the way they do
  - solve problems better
  - think "in parallel"
  - think critically → question and analyze everything honestly and scientifically
  - …

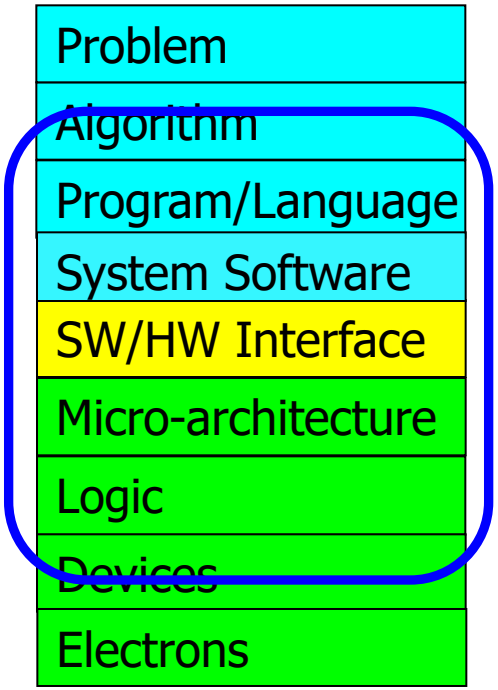
A Note on Hardware vs. Software

- This course might seem like it is only "Computer Hardware"

- However, you will be much more capable if you master both hardware and software (and the interface between them)
  - Can develop better software if you understand the hardware
  - Can design better hardware if you understand the software
  - Can design a better computing system if you understand both

The Transformation Hierarchy

Computer Architecture (expanded view)



What Do I Expect From You?

- Required background: Digital circuits course, programming, an open mind willing to take in many exciting concepts

- Learn the material thoroughly
  - attend lectures, do the readings, do the exercises, do the labs
- Work hard: this will be a hard, but fun & informative course
- Ask questions, take notes, participate
- Perform the assigned readings
- **Participate online (lecture, Moodle)**
- Start early
- If you want feedback, come to office hours



- Remember "Chance favors the prepared mind." (Pasteur)

What Do I Expect From You?

- How you prepare and manage your time is very important

- There will be many lab and homework assignments
  - They will take time
  - Start early, work hard

- This will be a heavy course
  - However, you will learn a lot of fascinating topics and critically analyze tradeoffs and make tradeoff-based decisions in computer system design
  - And, it will hopefully change how you look at and think about designs around you

How Will You Be Evaluated?

- Lab assignments: 50%
- Final exam (180 minutes): 30%
- Homeworks: 20%

- Many extra credit possibilities in HWs, Labs, Exam

Course Website

- https://safari.ethz.ch/architecture/fall2025/

- All slides, lecture videos, readings, assignments to be posted
- Plus, other useful information for the course

- Check frequently for announcements and due dates

Office Hours

- Goal: Answer questions about logistics and assignments
- Monday, 13.00-14.00 pm
  - Starting September 29
- Remote-only
- Check Moodle for zoom link

Lab Sessions

- Goal: Support and guidance for the Lab assignments

- Tuesday, 17.00-18.00 pm
  - Starting September 30

- Hybrid Sessions
  - In-person, ETZ H71
  - Online on Zoom
    - Check Moodle for the link



Homework 0

- Is due October 3
  - https://safari.ethz.ch/architecture/fall2025/lib/exe/fetch.php?media=hw0_ca_fall2025.pdf
- Information about yourself
- All future grading is predicated on homework 0

Heads Up

- Lab 1 is already out
  - Due on Oct. 10
- HW1 will be out on Oct. 3
  - Due in ~2 weeks after release
- Check the website and your email frequently for assignments and announcements
  - https://safari.ethz.ch/architecture/fall2025/
- My goal is to enable your learning and growth, so labs can be done any time until the end of the semester (Jan 31st)
  - But, please know yourself and plan accordingly

Required Reading

Richard Hamming

``You and Your Research''

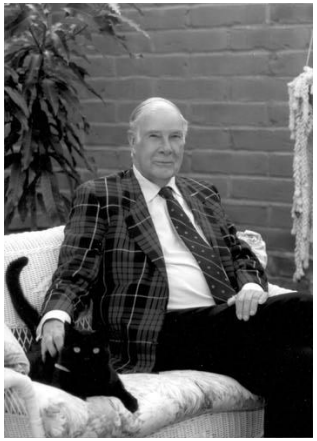
Transcription of the
Bell Communications Research Colloquium Seminar
7 March 1986

https://safari.ethz.ch/architecture/fall2024/lib/exe/fetch.php?media=youandyourresearch.pdf

Required Reading on Mindset & More

If you really want to be a first-class scientist you need to know yourself, your weaknesses, your strengths, and your bad faults, like my egotism. How can you convert a fault to an asset? How can you convert a situation where you haven't got enough manpower to move into a direction when that's exactly what you need to do? I say again that I have seen, as I studied the history, the successful scientist changed the viewpoint and what was a defect became an asset.

In summary, I claim that some of the reasons why so many people who have greatness within their grasp don't succeed are: they don't work on important problems, they don't become emotionally involved, they don't try and change what is difficult to some other situation which is easily done but is still important, and they keep giving themselves alibis why they don't. They keep saying that it is a matter of luck. I've told you how easy it is; furthermore I've told you how to reform. Therefore, go forth and become great scientists!



https://safari.ethz.ch/architecture/fall2024/lib/exe/fetch.php?media=youandyourresearch.pdf

Suggested Talk on Mindset & Research



https://www.youtube.com/watch?v=83tlorht7Mc&list=PL5Q2soXY2Zi8D_5MGV6EnXEJHnV2YFBJI&index=54

An Interview on Computing Futures



https://www.youtube.com/watch?v=8ffSEKZhmvo

A Recent Interview on Impactful Research



Building the Future of Robust Computing Systems: Interview with Onur Mutlu

Posted on **September 24, 2025** by **ewent**

Read the latest interview with Onur Mutlu in the IEEE Computer Society INSIDER Tech news on receiving the 2025 Harry H. Goode Memorial Award for his seminal contributions to computer architecture research and practice, especially in memory systems. He discusses his work on RowHammer and research that bridges the gap between academia and industry, while taking a look towards building the future of robust computing systems.

Interview: Building the Future of Robust Computing Systems, IEEE Computer Society Team, INSIDER news, 09/23/2025

SAFARI News: Onur Mutlu receives the 2025 Harry H. Goode Memorial Award

Home / Publications / Tech News / INSIDER

Building the Future of Robust Computing Systems

IEEE Computer Society Team

Published 09/23/2025

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An Interview with Dr. Onur Mutlu – 2025 Harry H. Goode Memorial Award Recipient



Dr. Onur Mutlu, Professor of Computer Science at ETH Zurich, has done research in computer architecture, computing systems, hardware security, memory and storage systems, and bioinformatics and is an innovator whose work has improved microprocessors and memory and storage systems used by billions of people.

Your identification of the RowHammer vulnerability has had a significant impact on hardware security and memory systems. What led you to this discovery, and how has it influenced subsequent research in memory systems?

RowHammer is a fascinating phenomenon that affects all modern main memory (i.e., DRAM) chips used in almost all computers today (comprising a more than 120 billion USD market size). It is the fact that repeatedly accessing one DRAM row causes bitflips in physically nearby DRAM rows that should not change at all, leading to data corruption. We have identified the phenomenon and analyzed it in detail for the first time in 2012. We have continued to study it since then, discovering new properties as well as new read disturbance mechanisms along the way, such as RowPress (in 2020-2023) and Variable Read Disturbance (in 2024-2025). We have come a long way, yet the phenomenon still fascinates me and there is a lot more to do in fundamentally understanding and very efficiently solving it, especially as memory technology scales to much denser capacities using increasingly smaller cells to store each bit of data. And, not only us, the phenomenon continues to fascinate researchers across hardware security, design automation, computer architecture, dependable systems, device analysis communities. Many works are published each year on understanding, analyzing, modeling the phenomenon as well as solving it using new and creative methods across the system stack (spanning both hardware and software). For example, top security and computer architecture conferences have been almost regularly having dedicated RowHammer sessions (sometimes multiple of them) since 2020.

https://safari.ethz.ch/building-the-future-of-robust-computing-systems-interview-with-onur-mutlu/

https://www.computer.org/publications/tech-news/insider-membership-news/build-robust-computing-onur-mutlu

SAFARI Introduction & Research

Computer architecture, HW/SW, systems, bioinformatics, security, memory



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Think BIG, Aim HIGH!

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https://www.youtube.com/watch?v=mV2OuB2djEs

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